Module 47

Transactions/2: Serializability

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March 18, 2026

Module Recap

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- Concurrent execution of transactions raise serializability issues that need to be addressed
- All schedules may not satisfy ACID properties

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- To introduce the notions of Serializability that ensure schedules for transactions that may run in concurrent fashion but still guarantee and serial behavior
- To analyze the conditions, called conflicts, that need to be honored to attain Serializable schedules

Module Outline

- Serializability

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- Conflict Serializability

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- Different forms of schedule equivalence give rise to the notions of:
  - a) Conflict Serializability
  - b) View Serializability

Recap Schedule 3: Serializable

- Let T1 and T2 be the transactions defined previously. The following schedule is not a serial schedule, but it is equivalent to Schedule 1

A	B	A+B	Transaction	Remarks
100	200	300	@ Start	
50	200	250	T1, write A	
45	200	245	T2, write A	
45	250	295	T1, write B	@ Commit
45	255	300	T2, write B	@Commit

Note: In schedules 1, 2 and 3, the sum 'A + B' is preserved

Recap Schedule 4: Not Serializable

- The following concurrent schedule does not preserve the sum of 'A + B'

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- We assume that transactions may perform arbitrary computations on data in local buffers in between reads and writes
- Our simplified schedules consist of only `read` and `write` instructions

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- Intuitively, a conflict between I_i and I_j forces a (logical) temporal order between them
- If I_i and I_j are consecutive in a schedule and they do not conflict, their results would remain the same even if they had been interchanged in the schedule

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- These swaps do not conflict as they work with different items (A or B) in different transactions

Conflict Serializability (3)

- Example of a schedule that is **not** conflict serializable:

T_3	T_4
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- We are unable to swap instructions in the above schedule to obtain either the serial schedule $\langle T_3, T_4 \rangle$, or the serial schedule $\langle T_4, T_3 \rangle$

Example: Bad Schedule

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- Schedule S is very bad! We withdrew $100 from account A, but somehow the database has recorded that our account now holds $201!

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- By first example, the outcome is the same as Serial schedule 1. But that's just a peculiarity of the data.
- So neither S nor T are serializable.

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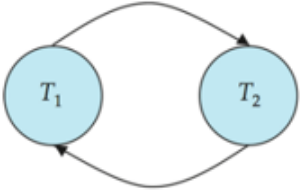

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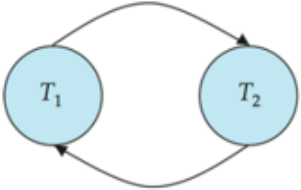

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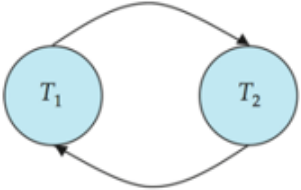

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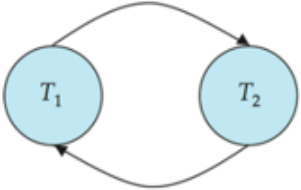

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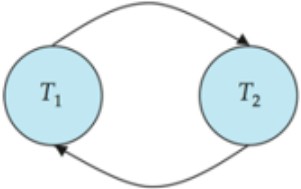
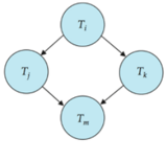
  - We may label the arc by the item that was accessed



Testing for Conflict Serializability

- A schedule is conflict serializable if and only if its precedence graph is **acyclic**



(a)

Testing for Conflict Serializability

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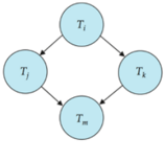
- Cycle-detection algorithms exist which take $\mathcal{O}(n^2)$ time, where n is the number of vertices in the graph



(a)

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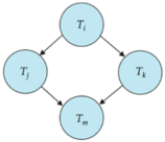

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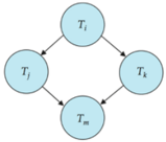

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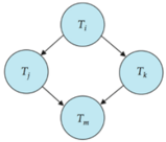

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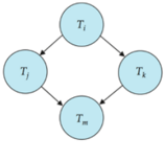
- If precedence graph is acyclic, the serializability order can be obtained by a **topological sorting** of the graph
  - That is, a linear order consistent with the partial order of the graph.
- For example, a serializability order for the schedule (a) would be one of either (b) or (c)



(a)

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- Go through each operation of the schedule.
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- If the operation is $r_i(X)$, find each subsequent write $w_j(X)$. Add a directed edge from T_i to T_j.
- The schedule is conflict-serializable if and only if the resulting directed graph is acyclic.
- Moreover, we can perform a topological sort on the graph to discover the serial schedule to which the schedule is conflict-equivalent.

Testing for Conflict Serializability (3)

Example

- Consider schedule: `w1(A), r2(A), w1(B), w3(C), r2(C), r4(B), w2(D), w4(E), r5(D), w5(E)`

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  - `w1(B)` $\rightarrow$ `r4(B)` $\Rightarrow T_1 \rightarrow T_4$

Testing for Conflict Serializability (3)

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  - `w2(D)` $\rightarrow$ `r5(D)` $\Rightarrow T_2 \rightarrow T_5$
  - `w4(E)` $\rightarrow$ `w5(E)` $\Rightarrow T_4 \rightarrow T_5$

- This graph has no cycles. One way to topologically sort is $T_3 - T_1 - T_4 - T_2 - T_5$. Serial schedule equivalent: `w3(C), w1(A), w1(B), r4(B), w4(E), r2(A), r2(C), w2(D), r5(D), w5(E)`

Module Summary

- Understood the issues that arise when two or more transactions work concurrently

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- Understood the issues that arise when two or more transactions work concurrently
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- Acyclic precedence graph can ensure conflict serializability

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